Laser printing tech produces waterproof e-textiles in minutes

Scientists from RMIT University have developed a cost-efficient and scalable method for rapidly fabricating textiles that are embedded with energy-storage devices. In just three minutes, the method can produce a 10cm x 10cm smart textile patch that is waterproof, stretchable and readily integrates with energy-harvesting technologies.

The technology enables graphene supercapacitors—powerful and long-lasting energy-storage devices that are easily combined with solar or other sources of power—to be laser-printed directly onto textiles.

In a proof-of-concept, the researchers connected the supercapacitor with a solar cell, delivering an efficient, washable and self-powering smart fabric that overcomes the key drawbacks of existing e-textile energy-storage technologies.

Dr Litty Thekkakara, researcher in RMIT's School of Science, says that smart textiles with built-in sensing, wireless communication or health monitoring technology called for robust and reliable energy solutions, as current approaches to smart textile energy storage, like stitching batteries into garments or using e-fibres, can be cumbersome and heavy, and can also have capacity issues.

She adds, "These electronic components can also suffer short-circuits and mechanical failure when they come into contact with sweat or with moisture from the environment."

This graphene-based supercapacitor is not only fully washable, it can store the energy needed to power an intelligent garment—and it can be made in minutes at large scale.

Water-filled Inmergo headphones allow immersive listening

Water rather than air carries sound in designer Rocco Giovannoni's Inmergo headphones, which promise rich audio even for people who are hard of hearing. A 2019 graduate of Royal College of Art's MA in Design Products, Giovannoni has designed the soft silicone headphones to improve upon the current bone-conduction audio technology.

Like other bone-conduction headphones, his design bypasses the eardrum and conveys sound as vibrations through the bones of the skull, directly to the cochlea. It does so via waterproof speakers that are fully immersed in liquid and then sealed in a pliable membrane. This membrane sits against a person's skin, transmitting sound vibrations through touch.

While existing bone-conduction headphones are popular among people who are deaf or have some hearing loss, and cyclists who want to listen to music but also remain aware of their surroundings, the devices are often hampered by poor sound quality.

Giovannoni says, "Inmergo aims to unravel the undiscovered potential of bone conduction by delivering innovative and inclusive sound experiences that can improve the quality of life."

Giovannoni has two functioning prototype models—a two-unit set of headphones and a five-unit helmet that wraps more comprehensively around the head. The units are filled with an ultrasonic gel, which the designer chose as the best-sounding fluid after many rounds of testing. Both models deliver bass frequencies more efficiently than any air-based design.

Immergo is designed with soft silicone and filled with water. The design creates vibrations, which travel through the bones of the skull (Credit: Mashable)

Giovannoni believes the technology could one day be used in hi-fi headphones for hearing loss, virtual reality (VR), sound- and thermo-therapy, and as a bass supplement to standard earphones.

e-Scooter that fits in a backpack

MiniFalcon scooter in its specially-designed backpack (Credit: Indiegogo)

MiniFalcon scooter has been created to be taken anywhere. It can be folded up in a specially-designed backpack, and maintains extremely high performance, long battery life and has a lot of safety features.

MiniFalcon features a kinetic energy recovery system, which recharges the battery when going downhill and during braking, keeping the rider on the road longer. A durable and lightweight aerospace aluminium body ensures a reliable ride with shortened brake time. Whereas, the anti-lock brake system allows the rider to maintain steering control when an obstacle appears.

Such features as powerful torque, adjustable gears, shock absorption and long-lasting battery make MiniFalcon the most portable e-scooter in the market.